

LogStruct: learned prior-anchored gene-graph smoothing for interpretable omics prediction

Draft status: active empirical draft. The code audit, METABRIC 1,000-gene 5-fold run, and one-fold Norman perturbation pilot are backed by logged runs under `results/`. The full 5k-10k feature campaign, remaining datasets, and biological validation are not complete.

Abstract

High-dimensional omics prediction often benefits from biological prior knowledge, but existing network-regularized linear models typically treat the gene graph as fixed. We evaluate LogStruct, a sklearn-compatible PyTorch estimator that learns a Bernoulli-parameterized feature graph anchored to a biological prior, uses the learned graph as an explicit feature-smoothing operator, and retains linear-model interpretability through effective coefficients. This draft currently reports an initial METABRIC breast-cancer run and a single-fold Norman perturb-seq pilot. These first results are mixed: LogStruct is competitive with a fixed STRING graph on METABRIC but trails tuned L2 logistic regression; on the Norman pilot, LogStruct improves macro-F1 over the tuned logistic baseline. These results are logged but not yet publication-complete.

1. Introduction

Genomic and single-cell prediction problems often have many more molecular features than independently sampled biological units. Sparse linear models are attractive in this regime because they are simple, reproducible, and inspectable, but ordinary lasso, ridge, and elastic net ignore curated knowledge about gene interactions, pathways, and regulatory structure. Fixed-graph Laplacian methods address this by encouraging neighboring genes to have similar coefficients, but the prior graph is often noisy, incomplete, or mismatched to the phenotype.

LogStruct tests a middle ground. It keeps the estimator linear after a learned feature-smoothing operator, while allowing the gene graph to deviate from a prior through a KL-anchored Bernoulli adjacency. This is designed for two regimes: bulk transcriptomics, where sample size is small and prior structure may improve sample efficiency, and single-cell data, where rare cell types or perturbations have limited support even when the total number of cells is large.

The planned contributions are:

1. A precise implementation audit of LogStruct’s smoothing, Laplacian, and KL terms, including effective coefficients for interpretation.
2. A controlled empirical comparison against fixed-graph Laplacian models, GELnet-style penalties, classical baselines, gradient boosting, MLPs, and GNNs.

3. Biological validation of learned effective coefficients and graph changes using pathway enrichment, marker recovery, gained/lost edges, and regulatory recovery in perturb-seq.

2. Related work

See `paper/related_work.md`. In brief, LogStruct is positioned between fixed-graph sparse linear methods such as Li and Li, Logit-Lapnet, glmgraph, and GELnet, and learned-graph neural methods such as DIAL-GNN. It should be framed as an incremental but testable combination rather than a new modeling paradigm.

3. Method

Let X in $\mathbb{R}^{(n \times p)}$ be a sample-by-feature matrix and let A_0 in $[0,1]^{(p \times p)}$ be a biological prior adjacency. LogStruct learns adjacency logits L and defines:

```
A = zero_diag(0.5 * (sigmoid(L / T) + sigmoid(L / T)^T))
S(A) = alpha I + beta rownorm(A)
Y_hat = X S(A) W^T + b
```

The objective for classification is:

```
CE(Y, Y_hat)
+ lambda_en EN(W)
+ lambda_smooth trace(W L_A W^T)
+ lambda_kl KL(q(A) || p(A0))
```

Regression replaces cross-entropy with mean squared error. $EN(W)$ is the elastic-net penalty. $L_A = D_A - A$ is the graph Laplacian. The KL term is a Bernoulli KL over adjacency probabilities. This branch adds an `offdiag_mean` KL reduction to avoid $O(p^2)$ scaling of the penalty.

For interpretation, the raw head weights W are not the coefficients on the original feature space. The effective coefficients are:

```
W_effective = W S(A)^T
```

All biological interpretation in this paper must use `effective_coef_`, not `raw_coef_`.

4. Experimental setup

Planned datasets:

- METABRIC PAM50 breast cancer subtype classification.
- TCGA pan-cancer classification, using the UCI subset for fast shakedown and UCSC Xena for the full 33-class run.
- GDSC drug-response regression.

- GTEx broad tissue classification.
- Tabula Sapiens immune-compartment cell-type classification with donor-held-out splits.
- Norman 2019 perturb-seq perturbation identity classification and regulatory recovery.

All experiments will use fixed train/validation/test splits, train-only preprocessing, bootstrap confidence intervals, paired significance tests, and per-fold prediction logs. No result enters this manuscript unless it appears in a `results.json` file with seed, hyperparameters, git SHA, wall-clock time, and hardware record.

5. Results

5.1 Code audit and smoke test

The code audit identified and fixed three implementation issues: missing effective coefficients, temperature-distorted prior probabilities in the KL anchor, and missing STRING alias resolution. It also added unthresholded `adjacency_raw_` to distinguish the prediction-time graph from the thresholded interpretation graph.

The synthetic smoke test in `results/smoke/results.json` passed with balanced accuracy 0.773, macro-F1 0.777, and OvR AUROC 0.965. This verifies the training pipeline only and is not a biological result.

5.2 Initial METABRIC comparison

The first real-data run uses cBioPortal METABRIC mRNA z-scores with CLAUDIN_SUBTYPE labels, keeping LumA, LumB, Her2, Basal, and Normal samples and excluding claudin-low, missing, and NC labels. The processed matrix contains 1,756 samples and 20,385 unique gene symbols. Each outer fold selects the top 1,000 genes by variance on the train/validation partition only. Results are logged in `results/metabric_pam50/results.json`; the table is regenerated from that file by `experiments/01_brca_pam50/make_outputs.py`.

Table: `paper/tables/table1_metabric_1000g.md`.

Key result: tuned L2 logistic regression is best in this first run with balanced accuracy 0.696 (95% bootstrap CI 0.674, 0.715). LogStruct STRING-700 and the frozen fixed-graph STRING-700 baseline are nearly tied at 0.670 balanced accuracy. This does not support a performance advantage for learned adjacency on this reduced METABRIC setting. It does show that the graph-smoothed linear model is competitive but currently weaker than a strong classical baseline.

5.3 Prior sensitivity

Pending real-data runs.

5.4 Frozen versus learned adjacency

Pending real-data runs.

5.5 Smoothing-operator ablation

Pending real-data runs.

5.6 Sample efficiency

Pending real-data runs.

5.7 Single-cell cell-type classification

Pending real-data runs.

5.8 Perturbation identity and regulatory recovery

Norman 2019 data were downloaded from Figshare article 24688110 and loaded from the labeled H5AD. This file is a prefiltered 27,658-cell by 2,000-gene object. For the first closed-set pilot, double perturbations were excluded and controls plus single-gene perturbations with at least 50 cells were retained, yielding 15,216 cells, 18 classes, and 2,000 genes. One outer fold was run with the top 500 train-selected genes. Because only one fold is complete, the confidence intervals in `paper/tables/table_norman_500g_pilot.md` are degenerate and should not be interpreted as final uncertainty estimates.

In this pilot, LogStruct identity reached balanced accuracy 0.559 and macro-F1 0.598, LogStruct STRING-700 reached balanced accuracy 0.554 and macro-F1 0.595, and tuned L2 logistic regression reached balanced accuracy 0.541 and macro-F1 0.467. This is promising enough to justify the full 5-fold perturbation run, but it is not yet the final perturb-seq result.

5.9 Biological interpretation

Pending real-data runs.

6. Discussion

The audit already clarifies two important limitations. First, dense adjacency learning materializes $p \times p$ matrices and therefore has an $O(p^2)$ memory ceiling. Second, KL scaling is not a minor numerical detail: the legacy summed KL can dominate at large feature counts, so the paper must report the chosen KL normalization and sensitivity to `lambda_kl`.

The first METABRIC result is a useful negative control for the paper’s honesty standard: LogStruct did not beat a tuned L2 logistic baseline, and the learned STRING graph barely differed from the frozen graph baseline at the tested settings. The likely explanations are the reduced 1,000-gene feature set, short

early-stopped neural optimization, and a KL/smoothness setting that may not move the adjacency enough. The next experiment should sweep `lambda_kl`, `lambda_smooth`, and temperature before making any claim about adaptive graph benefit.

The Norman pilot points in the opposite direction: with imbalanced perturbation classes, the smoothed linear model improved macro-F1 over tuned logistic regression on the first fold. This must be confirmed over the remaining outer folds and with DoRothEA priors before it becomes a paper claim.

7. Conclusion

Pending real-data runs.

Availability

Code branch: `experiments/v1`. A release, Zenodo DOI, and reproducible Table 1 notebook remain pending.